Q1)

a. Why is it necessary to include non-linearity in a neural network?

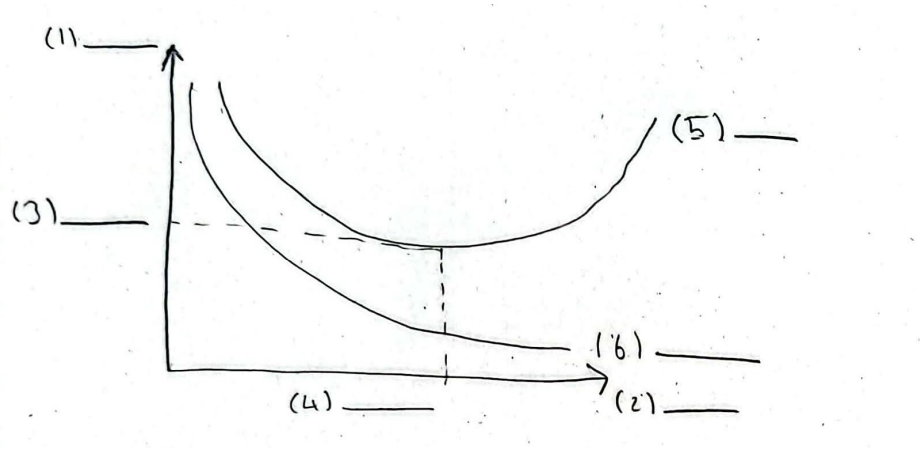
b. How do you check for overfitting in a neural network if your input data is 10-dimensional.

c. You are training a logistic regression model. You initialize the parameters with 0's. Is this a good idea? Explain your answer.

d. You design a fully connected NN architecture where all activations are sigmoids. You initialize the weights with large positive numbers. Is this a good idea? Explain your answer.

Q2)

Use the figure below to explain the concept of early stopping. Fill in the blanks (1) and (2), describe the axes (3) and (4), describe the values on the vertical and horizontal axis (5) and (6), describe the curves.



Q3)

Give the definitions of the following activation functions. Discuss their advantages and disadvantages. Indicate when these activation functions could be used.

a. Sigmoid
b. Tanh
c. ReLU
d. Leaky ReLU

Q4)

a. Describe the difference between the Sequential, Functional and Subclassing APIs of Keras. What are the advantages and disadvantages of them?
b. Consider the table given below. In each scenario (A, B, C), explain whether there is a high bias or high variance problem. Suggest solution to reduce that problem.

	Classification error (%)		
	A	**B**	**C**
Human error (proxy for Bayes)	1 0.7 0.5	1 0.7 0.5	0.5
Training error	5	1	0.7
Development error	6	5	0.8